ENGINEERING + DIGITAL TRANSFORMATION

PROJECT REFLOW

Reinventing a CAD $1.2 billion retail supply chain

CASE IDENTIFIER	CASE LEVEL	CHALLENGE FORMAT
AC-001 / Planning year 2027	Advanced	3 hours / team / 12-slide pitch

Client: NorthStar Retail Group (fictional) | Industry: Canadian retail and e-commerce | Planning horizon: 2027-2030

Case status: Original practice simulation inspired by consulting and engineering case competitions. No affiliation with, endorsement by, or official case materials from Accenture.

1. The situation

NorthStar Retail Group operates an e-commerce distribution network serving millions of Canadian customers. Despite growing sales, the company faces declining operational efficiency, rising logistics costs, inconsistent delivery reliability and pressure to reduce greenhouse-gas emissions.

The CEO believes artificial intelligence, warehouse automation and intelligent transport systems may help. The board has approved a maximum CAD $20 million upfront investment. Your team must recommend where the company should invest and how to execute the transformation.

2. Current operating performance

METRIC	BASELINE
Annual revenue	CAD $1.2 billion
Annual orders processed	12 million
Fulfillment and delivery cost / order	CAD $7.20
Annual logistics operating cost	CAD $86.4 million
Late delivery rate	8.5%
Fulfillment-related return/error rate	2.4%
Logistics greenhouse-gas emissions	18,000 tonnes CO2e / year
Distribution centres	3
Available capital investment	CAD $20 million

Demand is expected to grow, but the board has not adopted an official order-growth forecast.

3. Transformation options

Option A - AI-powered warehouse automation

Robotic picking assistance, computer vision, predictive maintenance and smart inventory positioning.

MEASURE	VALUE
Upfront investment	CAD $8.0M
Annual gross savings	CAD $6.2M
Added annual operating expense	CAD $1.3M
Late-delivery improvement	2.0 percentage points
Emissions reduction	8%
Implementation	6 months

Risks: Workforce adoption, robotics integration, downtime and maintenance complexity.

Option B - Intelligent transportation optimization

Machine-learning route optimization, shipment consolidation, delivery scheduling and vehicle utilization.

MEASURE	VALUE
Upfront investment	CAD $5.0M
Annual gross savings	CAD $4.8M
Added annual operating expense	CAD $0.9M
Late-delivery improvement	3.2 percentage points
Emissions reduction	18%
Implementation	4 months

Risks: Data accuracy, algorithm reliability, carrier integration and cybersecurity.

Option C - Regional micro-fulfillment network

Smaller automated facilities located closer to customer demand.

MEASURE	VALUE
Upfront investment	CAD $12.0M
Annual gross savings	CAD $7.0M
Added annual operating expense	CAD $2.2M

MEASURE	VALUE
Late-delivery improvement	2.5 percentage points
Emissions reduction	12%
Implementation	12 months

Risks: Facility location, real estate, demand uncertainty and operational complexity.

4. Portfolio and financial modelling rules

You may select one or more initiatives provided total upfront investment does not exceed CAD $20 million. Benefits are not perfectly independent.

• A + B: Deduct 15% of the smaller initiative's annual gross savings from combined gross savings.

• A + C: Deduct 30% of the smaller initiative's annual gross savings.

• B + C: Deduct 10% of the smaller initiative's annual gross savings.

• All three: Apply pairwise overlap deductions; however, this portfolio exceeds the capital budget.

• Added annual operating expenses are additive.

• Late-delivery improvements are additive, capped at a total improvement of 5.0 percentage points.

• Emissions reductions are additive, capped at 25% overall.

These are simplified educational assumptions, not empirical engineering forecasts.

5. Board decision constraints

• Capital expenditure must remain at or below CAD $20 million.

• Reduce late deliveries from 8.5% to 4.5% or less.

• Reduce annual logistics emissions by at least 20%.

• Create positive financial value over the specified evaluation period.

• Protect customer data and minimize operational disruption.

• Support increasing annual order volume.

6. Financial evaluation instructions

Use a 10% annual discount rate. All investments occur at the start of Year 0. Net annual savings occur at the end of Years 1, 2 and 3. Ignore taxes, inflation, terminal value and residual asset value in the base case.

Net annual savings = total gross savings - overlap deductions - additional annual operating expenses.

Three-year NPV = - upfront investment + sum of discounted net savings for Years 1-3. State your exact ROI definition and simple-payback assumptions.

7. CEO question

“How should NorthStar invest its $20 million to build a more efficient, profitable, reliable and sustainable logistics network?”

8. Required analysis

A. Root cause - Explain the operational problems and why they matter.

B. Engineering - Assess warehouse throughput, transport efficiency, capacity, reliability and scalability.

C. Alternatives - Compare every budget-feasible single and combined investment portfolio.

D. Financial modelling - Calculate annual net savings, ROI, simple payback and three-year NPV for each portfolio.

E. Sustainability - Estimate tonnes of CO2e avoided and whether the target is reached.

F. Implementation - Plan phased rollout, milestones, owners, KPIs and continuity safeguards.

G. Innovation - Suggest feasible additional improvements; distinguish quantified benefits from hypotheses.

H. Risk - Examine workforce, execution, cybersecurity, technology, financial and regulatory risks.

9. Required deliverables

• A 12-slide consulting presentation and one-page executive summary.

• A transparent financial model and portfolio comparison.

• A risk register with probability, impact, mitigation and accountable owners.

• A three-year implementation roadmap with measurable performance indicators.

• A clearly defended final recommendation and two-minute executive pitch.

• Responses to five difficult executive questions.

10. Judge challenge - prepare precise answers

1. Why is your portfolio better than the highest gross-savings alternative?

2. What if implementation costs increase by 25%?

3. What if projected savings are 30% below forecast?

4. How would the strategy change if order volume increased by 40%?

5. Why should NorthStar trust AI-driven systems with essential operations?

11. Evaluation rubric

CATEGORY	POINTS
Problem diagnosis	15
Engineering and technical analysis	20
Financial accuracy	20
Strategy and innovation	15
Implementation and risk	15
Presentation and executive defense	15
TOTAL	100

Final instruction: Do not recommend AI merely because it is fashionable. Demonstrate measurable value, adherence to constraints, and an implementable plan. This packet intentionally excludes the reference solution and answer key so that the AI or competition team can be independently evaluated.